APPLICATION REPORT · V1.2.0

MedMap Visual Atlas

An AI-assisted clinical education platform: symptom-to-mechanism mapping, emergency case simulation, pharmacology interaction screening, knowledge-graph mind maps and board-style quizzes, with a built-in publish-readiness gate.

REPORT DATE	CODE VERSION	REPOSITORY
12 November 2026	Commit `14cea8f` · package 1.2.0	Subham2/Medmap-visual-atlas
STATUS	**RUNTIME**	**AI PROVIDER (TESTED)**
Draft PR #1 · CI passing · no conflicts	Node.js ≥ 22.19 · Express 4 · React 19	aicredits.in (OpenAI-compatible)

> **Medical disclaimer**
> MedMap Visual Atlas is for education and simulation only. It does not provide medical advice, diagnosis or treatment. AI-generated content can be wrong; verify all clinical, diagnostic, therapeutic and drug information against current guidelines and qualified professionals.

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1 Executive summary

The uploaded codebase imported 14 modules that did not exist, carried a live provider API key in source, and in practice served only offline text. It has been completed, debugged, optimized and tested end to end against the real provider. Every AI feature now returns live generated content, and a 33-check publish gate runs locally and in CI.

33/33
Publish-gate checks passing (offline mode)

8/8
AI content routes returning live output in `--live` mode

34 → 0
Production dependency vulnerabilities

493 → 152 KB
Firebase bundle (lazy Firestore)

~20 s
10 AI cases or a full 21-section mind map

0
Console errors in the live browser run

What made it "a static collection of texts"

Five independent faults each caused a silent switch to offline data:

1. **Proxy ignored.** Node's built-in `fetch` does not read `HTTPS_PROXY`; every provider call failed with `ENOTFOUND`.
2. **Gateway time cap.** The gateway terminates requests at about 30 s. Ten cases or a multi-thousand-word mind map in one call cannot finish in time.
3. **Concurrency cap.** The key allows 5 simultaneous requests; bursts returned HTTP 429 "Concurrency Limit Exceeded".
4. **Slow default model.** `gemini-3.5-flash` failed most large chunks inside the cap.
5. **Invented image URLs.** The model fabricated plausible Wikimedia file names; the verifier rejected every one as a 404.

> **Security action required**
> Two provider keys must be rotated in the provider dashboard: the key that was hard-coded in the original `server.ts` (prefix `sk-live-c785…`) and the key shared in the development conversation. Neither is in any commit; the current key exists only in a git-ignored local `.env`.

2 Feature modules

MODULE	WHAT THE LEARNER DOES	BACKEND
Symptom Mechanism Mapper left column, always visible	Pick symptoms and case details; get a 12-section map: overview, master table, pathophysiology, histopathology, biochemistry, genetics, anatomy/physiology, investigations, differentials, therapy, exam points and verified images. Applying a new focus topic rebuilds it with that topic's cardinal symptoms.	`/api/symptom-mechanism-map`, `/api/medical-images`
Emergency Case Simulator	Generate 10 case types for any topic (classic, subtle, severe, atypical, pediatric, elderly, pregnancy, comorbidity, complication, differential challenge). Order tests, commit to a diagnosis, choose a treatment and watch the vitals respond on a live ECG monitor.	`/api/cases`, `/api/generate-cases`
Patient monitor	Animated lead-II ECG whose rate follows heart rate, with an ST-elevation morphology after a correct STEMI diagnosis. Colour-coded alarm thresholds for HR, BP (with MAP), SpO_2 and temperature.	Client only (canvas)
Clinical journal and Sheets sync	Every run is logged locally with attending feedback and learning points; export to CSV, or sign in with Google to append to a "MedMap Clinical Simulation Logs" spreadsheet with AI grading.	`/api/sync-to-sheets`
Knowledge-graph mind map	21-section radial graph with cross-links; click a node to read its section; a therapeutics panel (guidelines, advanced and emerging therapies, algorithm, monitoring). Export as PNG, SVG, PDF, JSON, Markdown or Mermaid.	`/api/generate-mindmap`
Pharmacology matrix	Topic formulary, add any drug, then check pairs from dropdowns or by clicking cells in an N×N severity grid (results cached per pair).	`/api/get-topic-drugs`, `/api/get-drug-interaction`
Clinical quiz	Board-style vignettes with scoring, explanations and pearls. Malformed AI questions are filtered out before display.	`/api/clinical-quiz`
Audit & compliance	Live publish-readiness verdict and checklist, AI tier in use, simulation audit statistics, Sheets ledger status and a downloadable JSON report.	`/api/readiness`, `/api/status`

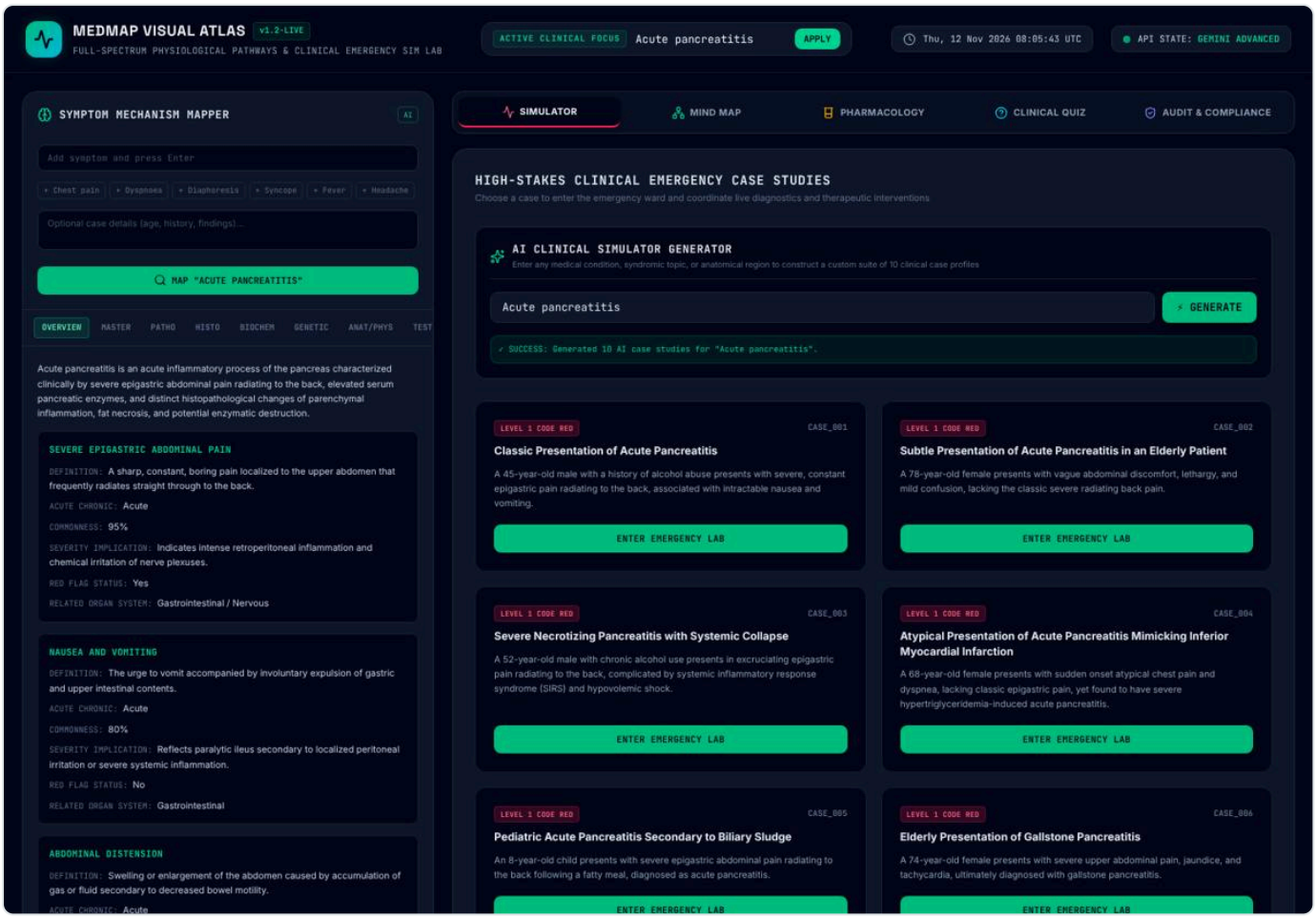


Figure 1. Live mode. Applying "Acute pancreatitis" rebuilds the symptom mapper with that topic's cardinal symptoms (here "Severe epigastric abdominal pain"); the header shows the live AI state.

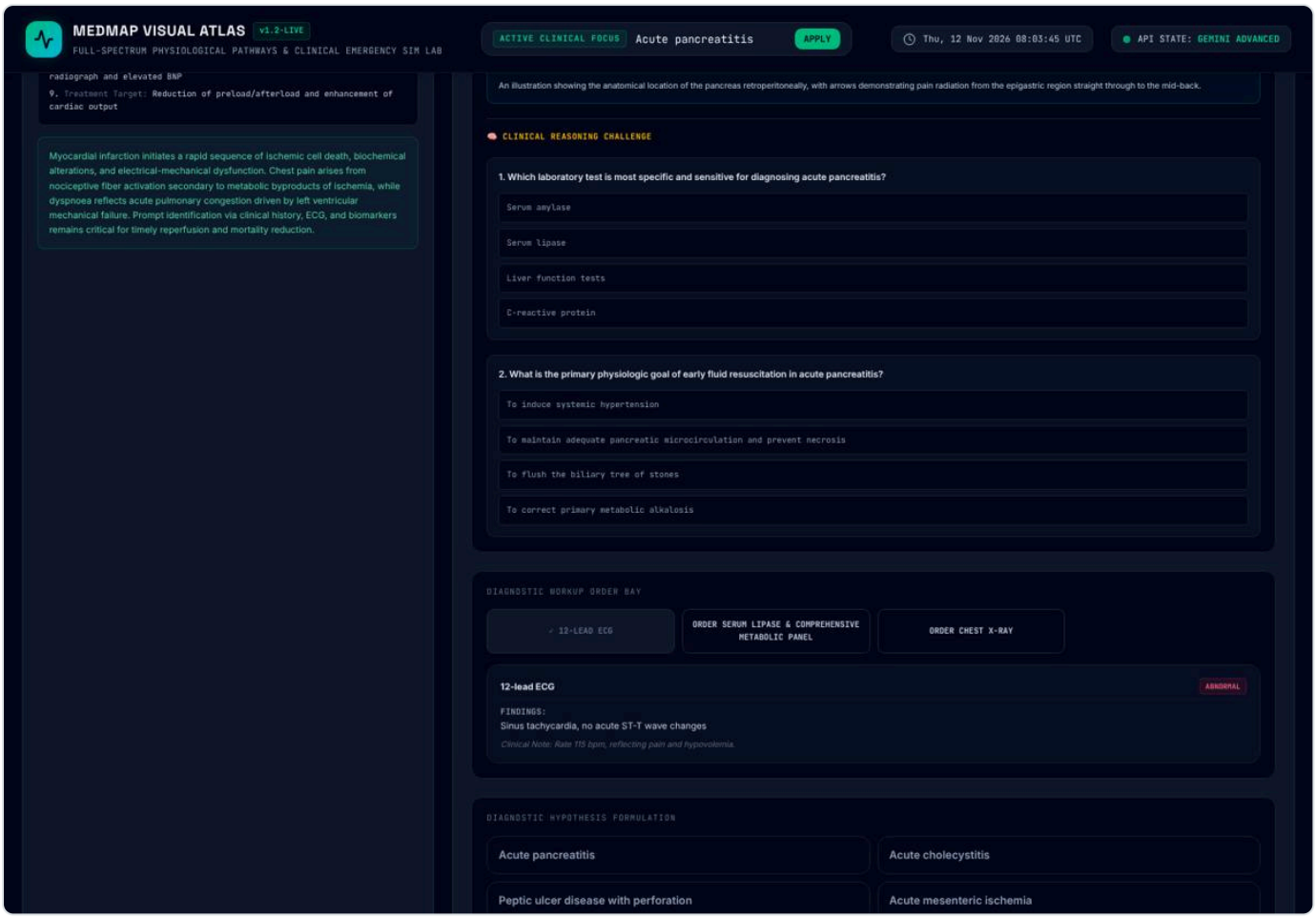


Figure 2. Case simulator with an AI-generated acute pancreatitis case: ordered ECG result (sinus tachycardia attributed to pain and hypovolaemia) and monitor.

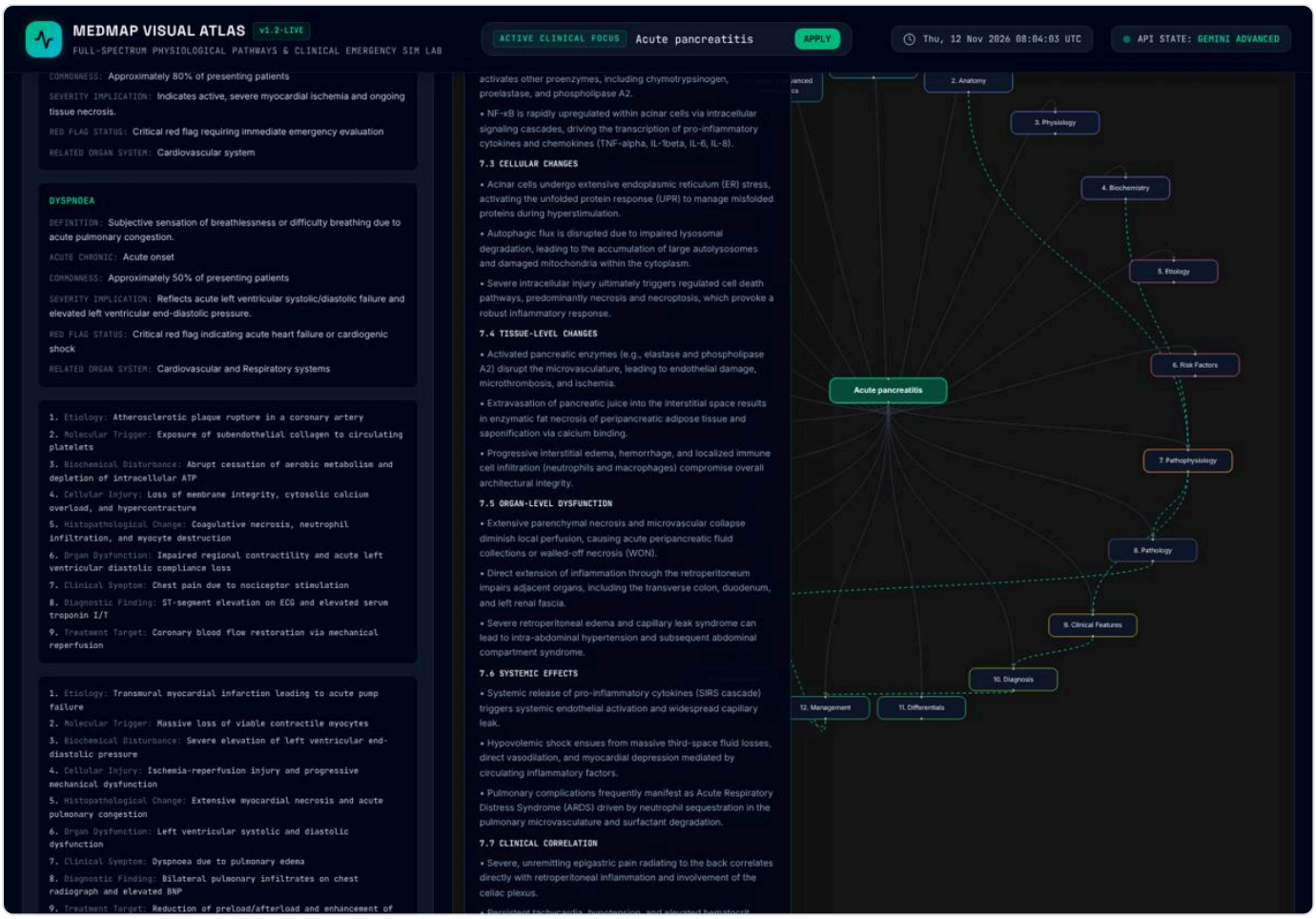


Figure 3. Live knowledge-graph mind map for acute pancreatitis with the Pathophysiology section open.

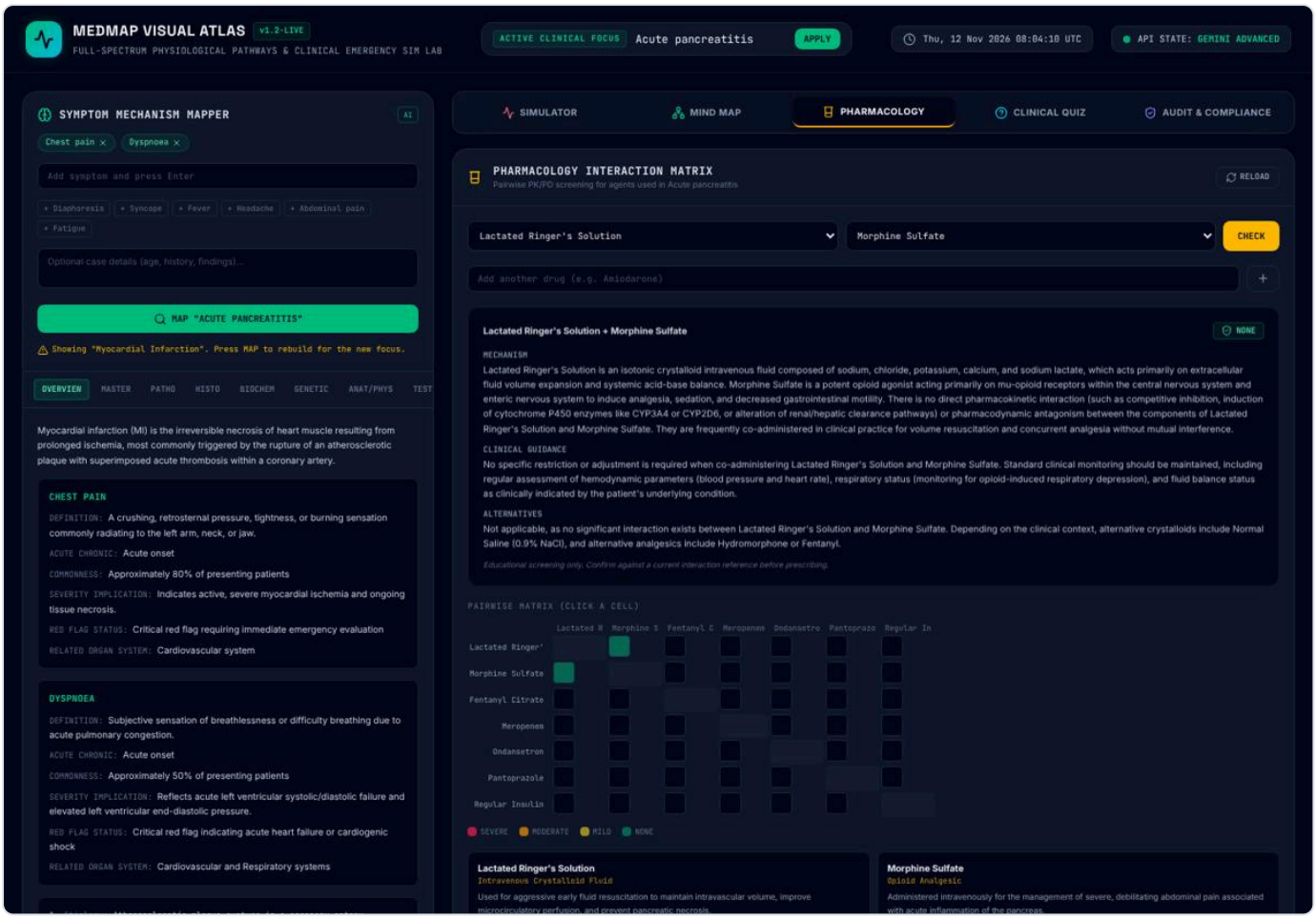


Figure 4. Pharmacology matrix: live formulary, interaction result and pairwise severity grid.

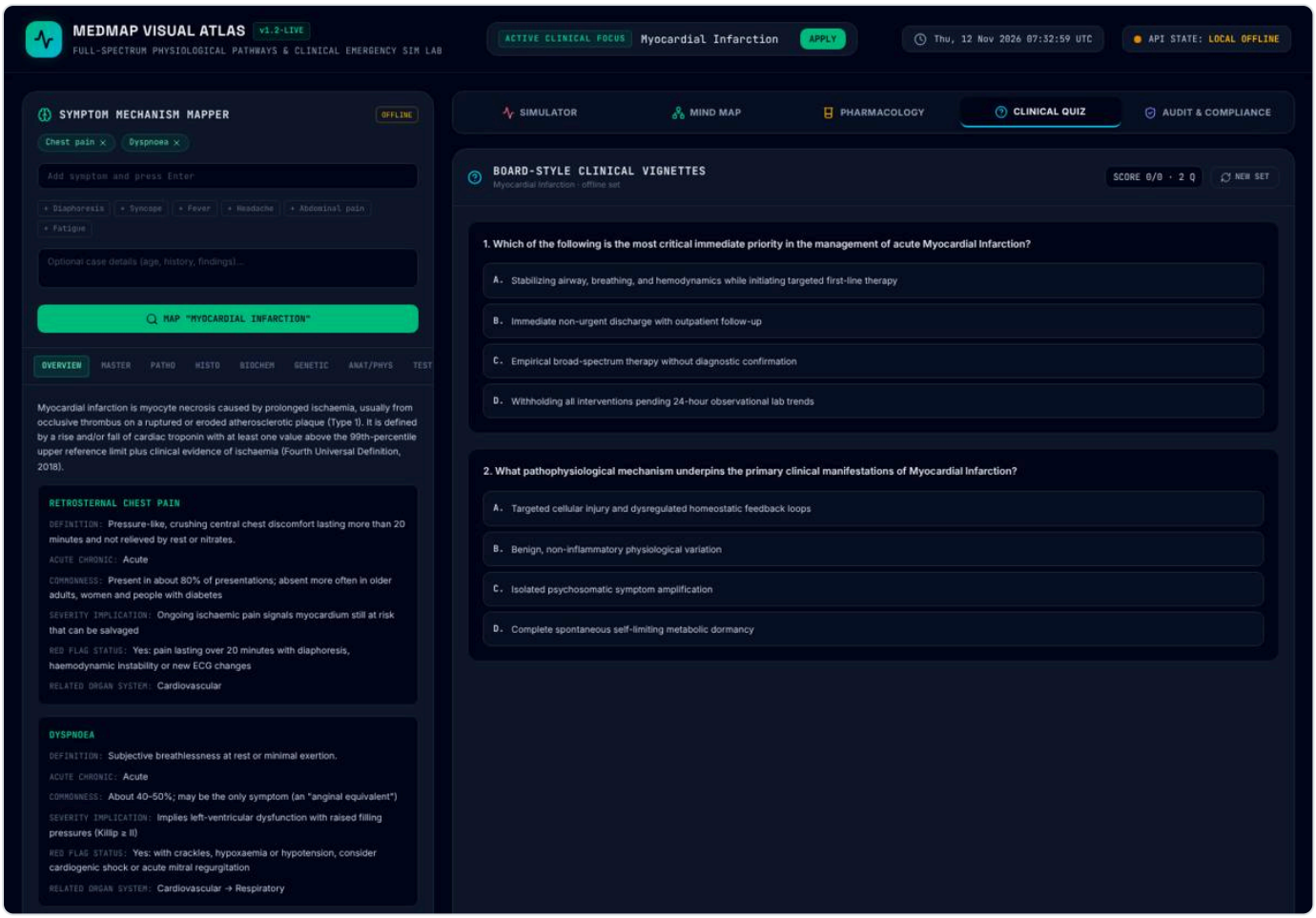


Figure 5. Clinical quiz tab with scoring.

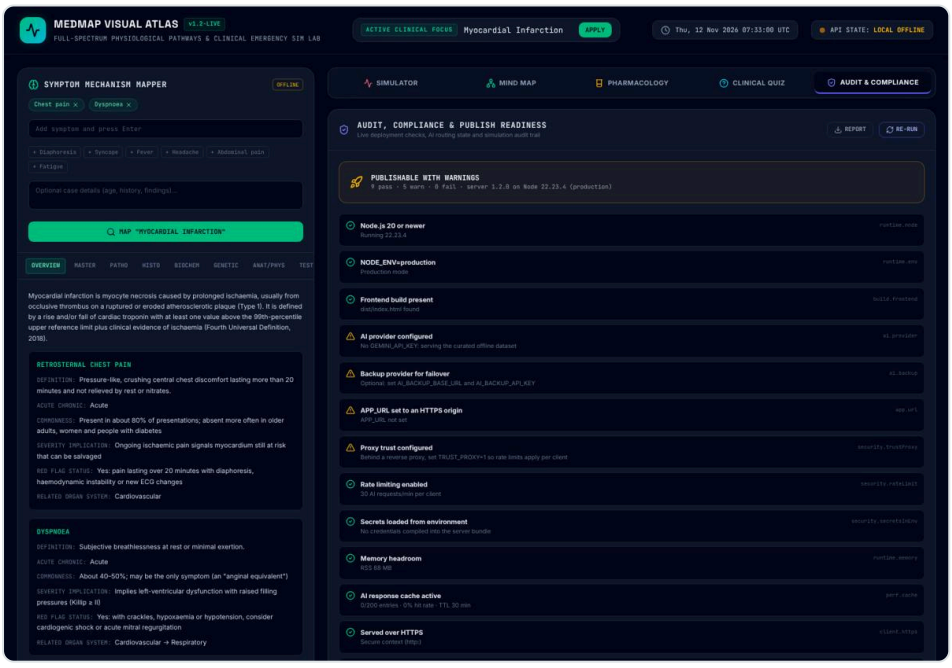


Figure 6. Audit & compliance tab running without an AI key: "Publishable with warnings", each check with its reason.

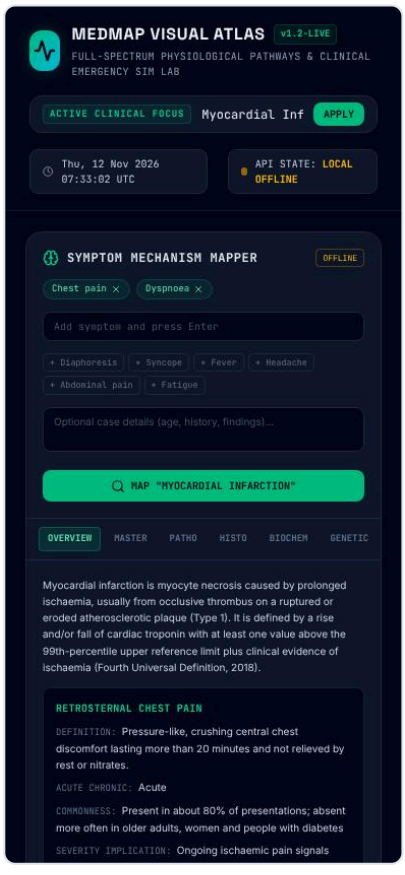


Figure 7. 390 px phone width: no horizontal overflow.

3 System architecture

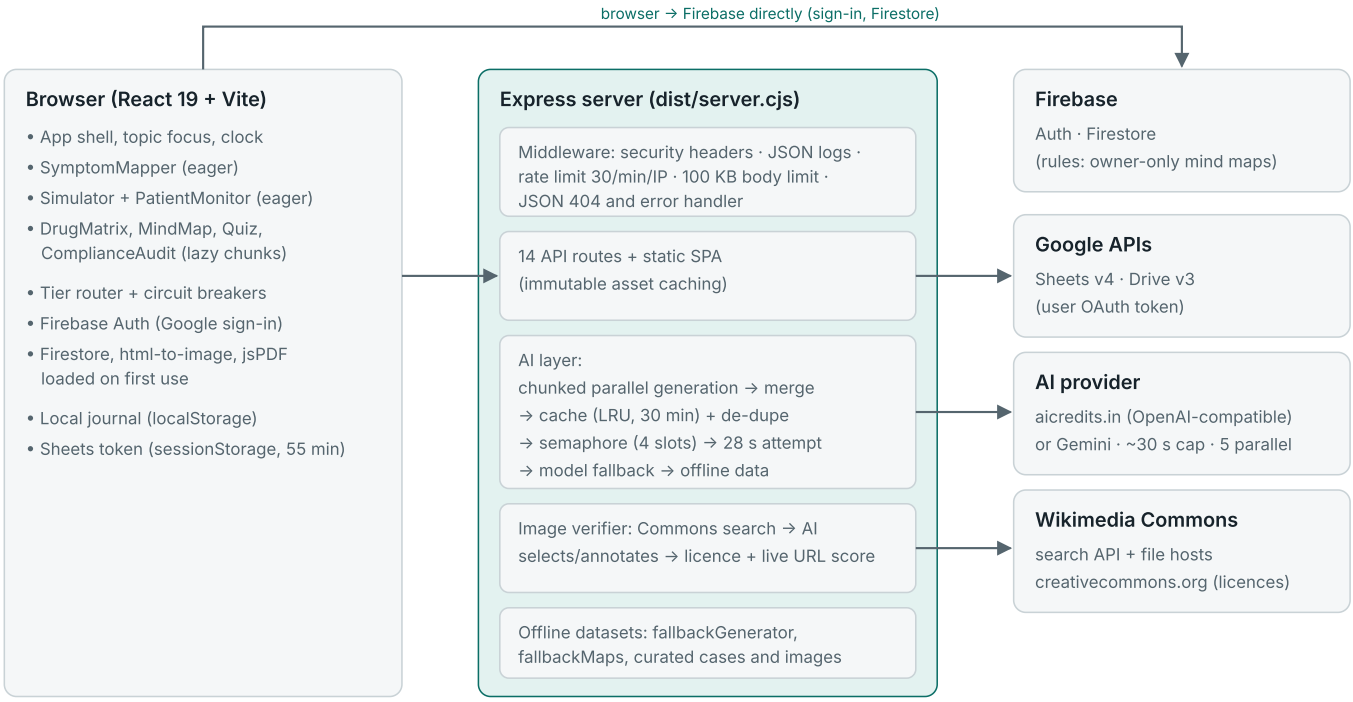


Figure 8. Runtime architecture. Each external service sits beside the server layer that calls it; arrows show request direction. Sheets requests go through the server with the user's OAuth token; Firebase is called directly from the browser.

Technology stack

LAYER	TECHNOLOGY	ROLE
Frontend	React 19, Vite 6, Tailwind CSS 4, lucide-react	UI, code splitting, styling
Graph	@xyflow/react 12 (React Flow)	Mind-map rendering
Export	html-to-image, jsPDF (dynamic import)	PNG / SVG / PDF export
Auth and data	Firebase 12 (Auth, Firestore)	Google sign-in; owner-scoped mind-map storage
Server	Express 4, esbuild bundle, dotenv	API, static hosting
AI SDK	@google/genai 2 + native fetch to `/chat/completions`	Gemini direct, or any OpenAI-compatible gateway
Networking	undici (lazy, only when a proxy is set)	Outbound proxy support
Tooling	TypeScript 5.8, tsx, Playwright (tests), GitHub Actions	Typecheck, dev server, verification, CI

4 Repository structure

40 tracked files, about 11,300 lines (excluding the lockfile). Files marked new were written during this engagement because the uploaded code imported them but they did not exist.

PATH	LINES	PURPOSE
`server.ts`	2,765	Express server: middleware, AI layer, 14 routes, readiness, static hosting
`fallbackGenerator.ts`	2,999	Offline mind maps (MI, aortic dissection, DKA, pneumothorax, anaemias, generic) and cardiac case set
`src/App.tsx`	1,490	Shell, simulator, journal, Sheets sync, tab routing (lazy tabs)
`src/components/SymptomMapper.tsx` new	343	Mechanism mapper and image atlas
`src/components/ObsidianMindMap.tsx` new	340	Knowledge graph, section drill-down, therapeutics, exports
`src/components/ComplianceAudit.tsx` new	286	Publish-readiness UI, tier status, audit trail
`src/components/DrugMatrix.tsx` new	243	Formulary, interaction checks, severity grid
`src/components/PatientMonitor.tsx` new	228	Canvas ECG and vitals alarms
`src/components/ClinicalQuiz.tsx` new	162	Vignette quiz with scoring
`src/data/fallbackMaps.ts` new	329	Offline symptom maps: curated MI, labelled template for other topics
`src/lib/firebase.ts`, `firebaseAuth.ts`, `sheetsService.ts` new	242	Firebase init (lazy Firestore), Google sign-in with scopes, Sheets/Drive helpers
`src/lib/api.ts` new	49	Fetch wrapper with timeout and abort
`src/lib/ai/*` (types, webgpuSniffer new; circuitBreaker, tierRouter)	249	Inference-tier routing and failure breakers
`src/types.ts`	304	Shared domain types (cases, vitals, maps, images)
`scripts/publish-check.mjs` new	306	Publish-readiness gate
`Dockerfile`, `.dockerignore`, `.github/workflows/ci.yml` new	50	Container build (Node 22, non-root, healthcheck) and CI
`.env.example`, `.gitignore` new	50	Configuration template; secrets and build output ignored

PATH	LINES	PURPOSE
`firebase-*.json`, `firestore.rules`	46	Firebase client config, data blueprint, owner-only security rules
`README.md`, `FINAL_DEPLOYMENT_SETUP.md`, `docs/deployment-setup.html`	616	Documentation (updated)

5 API reference

All responses are JSON. AI routes return `{ source, data, notice? }`, where `source` is `"gemini"` for live AI output, or a `fallback-*` value explaining why offline data was used. Text inputs are trimmed and length-capped before use in prompts.

METHOD · PATH	INPUT	OUTPUT	AI CALLS
`GET /api/health`	none	`{status, uptimeSec}` (liveness)	none
`GET /api/status`	none	`{configured, source, provider, model}`; never exposes secrets	none
`GET /api/readiness`	none	Checklist report; HTTP 503 when a blocking check fails	none
`GET /api/cases`	none	3 curated emergency cases (MI, stroke, PE)	none
`POST /api/generate-cases`	`{topic}`	Up to 10 validated, playable cases	5 parallel × 2 cases
`POST /api/symptom-mechanism-map`	`{topic, symptoms?, case_details?}`	Full `SymptomMechanismMap`	4 parallel key groups
`POST /api/generate-mindmap`	`{topic}`	`{topic, markdown, nodes, links, therapeutics}`	1 graph + 5 markdown + 2 therapeutics
`POST /api/analyze-symptoms`	`{symptoms: string[]}` (≤ 15)	4 differentials with probability, tests, pearl	1
`POST /api/get-drug-interaction`	`{drugA, drugB}`	Severity, mechanism, guidance, alternatives	1
`POST /api/get-topic-drugs`	`{topic}`	6–10 drugs with class and use	1

METHOD · PATH	INPUT	OUTPUT	AI CALLS
`POST /api/clinical-quiz`	`{topic}`	Validated vignette MCQs	1
`POST /api/medical-images`	`{topic}`	2–4 verified, licensed images with teaching notes	1 (plus Commons search)
`GET /api/wikimedia-search`	`?query=`	Proxied Commons search	none
`POST /api/sync-to-sheets`	`Authorization: Bearer <Google token>` + run fields	`{spreadsheetId, attendingFeedback, learningPoints, timestamp}`; 401 on expired token	1 (grading)

Errors: malformed JSON → 400, oversized body → 413, unknown `/api` route → 404, rate limit → 429 with `Retry-After`, unexpected → 500. All are JSON.

6 AI pipeline

6.1 Provider resolution

- No `GEMINI_API_KEY`: offline mode (curated data). Nothing is hard-coded.
- A Google key (`AIza…`): Gemini API directly via `@google/genai`.
- Any other key with `GEMINI_BASE_URL`: an OpenAI-compatible `/chat/completions` endpoint, with JSON response mode when structured output is needed.
- Optional `AI_BACKUP_BASE_URL` / `AI_BACKUP_API_KEY`: failover used only when the Gemini route fails.

6.2 Measured provider limits (aicredits.in)

PROPERTY	OBSERVED	DESIGN RESPONSE
Request time cap	HTTP 500 "Internal Server Error" at ≈ 30 s	Chunked generation; 28 s per-attempt timeout leaves time to fail over
Concurrency	5 parallel requests per key; the 6th returns 429	Process-wide semaphore, 4 slots; 429 waits and retries the same model (≤ 3 times)
Bad credentials	401 / 402 / 403	Stop immediately (1 upstream call instead of 4)
Malformed JSON	Occasional invalid escapes (e.g. LaTeX `\(`)	Parser repairs invalid escapes and control characters, then retries the parse

6.3 Model benchmark

All three heavy routes launched simultaneously (17 chunks through the 4-slot queue):

MODEL	10 CASES	SYMPTOM MAP	MIND MAP	CHUNKS FAILED
`gemini-3.5-flash`	53 s (4 cases)	90 s	153 s	9 / 17
`google/gemini-3.5-flash-lite` (selected)	**21 s (10 cases)**	**29 s**	**40 s**	**1 / 17**
`openai/gpt-4.1-mini` (fallback)	34 s (7 cases)	51 s	71 s	0 / 17

Single-call throughput measured directly: flash-lite ≈ 320 tokens/s, gpt-4.1-mini ≈ 120 tokens/s. Sequential live latencies after tuning: short answers 3–12 s, 10 cases ≈ 20 s, mind map ≈ 18–20 s, symptom map ≈ 6–8 s.

6.4 Chunked generation

ROUTE	CHUNKS	MERGE RULE
Cases	5 × 2 case types, ids pre-assigned per chunk	Concatenate; drop unplayable cases; re-key duplicate ids; any chunk may fail
Symptom map	4 key groups (overview/master; patho/histo/biochem; genetics/anatomy/investigations; differentials/therapy/exam/mind-map)	Overview group required; missing groups filled from offline map; images always from the verified pipeline
Mind map	Graph (nodes and links) + 5 markdown section ranges + 2 therapeutics groups	Graph required; missing markdown ranges or therapeutics filled from offline content

6.5 Caching and request flow

```
route → chunk prompts → callGeminiWithRetry
          ├─ cache hit (LRU, 200 entries, 30 min TTL)  → return
          ├─ identical request in flight               → share its promise
          └─ semaphore slot (4) → attempt (≤ 28 s)
                 ├─ 429 → wait 1–3 s, retry same model (≤ 3)
                 ├─ 401/402/403 → stop
                 └─ timeout/5xx → next model (AI_FALLBACK_MODELS)
       → validate/merge → { source: "gemini" | "fallback-…", data }
```

6.6 Image retrieval

1. Search Wikimedia Commons (file namespace, bitmap only) for the topic; keep JPEG/PNG/GIF/WebP files with CC or public-domain licences, excluding NC/ND.
2. The model selects 2–4 medically relevant candidates by index and writes teaching points and a pathology correlation. It never supplies URLs.
3. The verifier scores each image out of 100: source credibility 25, licence clarity 25, live URL and image content-type 20, medical relevance 15, metadata completeness 10, quality 5. ≥ 80 is verified, ≥ 60 needs review, otherwise rejected. It checks the original file and displays the thumbnail.
4. An identifying User-Agent, a per-topic result cache and a `Retry-After` cooldown comply with Wikimedia's rate-limit policy.

7 Offline mode and content integrity

Offline mode activates when no key is configured or a provider call fails. It is designed never to present invented, topic-specific "facts".

CONTENT	OFFLINE BEHAVIOUR
Cases	Cardiac topics use the 10-case cardiac generator; stroke and PE topics get their curated case; anything else gets the three curated cases. Previously, MI cases were relabelled for any topic (an "Asthma" case had ST-elevation ECGs).
Symptom map	Curated MI map; other topics get a structured reasoning template clearly labelled "OFFLINE TEMPLATE".
Mind maps	Curated content for MI, aortic dissection, DKA, pneumothorax, anaemias and sideroblastic anaemia; generic scaffold otherwise.
Images	Curated verified images only when the topic matches; otherwise none. Previously, unrelated images were shown as "verified".
Drug interaction, symptom analysis, quiz	Rule-based answers for common patterns; generic guidance otherwise.

Every response carries `source`; the UI shows AI or OFFLINE badges and a header state indicator.

8 Security and compliance

CONTROL	IMPLEMENTATION
Secrets	Loaded only from the environment; `.env` git-ignored; secret scan of source and the server bundle on every publish check and CI run. A hard-coded key in the original code was removed before the first commit.
Transport and headers	`X-Content-Type-Options: nosniff`, `Referrer-Policy`, `Permissions-Policy`, `Cross-Origin-Opener-Policy: same-origin-allow-popups` (for Google sign-in), HSTS in production, `X-Powered-By` removed
Abuse protection	Per-IP limit on AI POST routes (30/min default; `TRUST_PROXY` for real client IPs); 100 KB JSON body limit; prompt inputs length-capped
Google Sheets	Scopes limited to `spreadsheets` + `drive.file` (only files the app created); token in `sessionStorage` for 55 min; spreadsheet ID validated; `RAW` value input so cells are never evaluated as formulas; expired token returns 401 and prompts re-sign-in
Firestore	Rules deny everything except owner-only read and write of `/mindmaps/{id}`
Firebase web key	Public client identifier by design; restrict it by HTTP referrer in Google Cloud Console
Dependencies	Production audit: 34 vulnerabilities → 0 (lockfile fixes plus pinned `qs ≥ 6.17.1` and `@grpc/grpc-js ≥ 1.14.7`); CI fails on high severity
Container	Multi-stage build, production dependencies only, runs as non-root `node`, health check on `/api/health`
Medical disclaimer	In README (enforced by the gate), the compliance tab and therapeutics panels
Browser privacy	The Ollama localhost probe is opt-in, avoiding Chrome's local-network permission prompt on public pages

9 Performance and optimization

Frontend bundle (production build)

CHUNK	SIZE	GZIP	LOADED
App entry (React, app, symptom mapper, simulator)	255.4 KB	77.0 KB	Initial
Firebase (app + auth)	154.2 KB	31.6 KB	Initial
Stylesheet	58.9 KB	10.3 KB	Initial
React Flow	191.1 KB	62.7 KB	Mind-map tab
jsPDF + html2canvas + canvg + DOMPurify	782.8 KB	242.1 KB	First PDF/PNG/SVG export
Tab components (4)	39.1 KB	15.2 KB	On tab open

The publish gate measures initial JavaScript at 167 KB gzip against a 200 KB budget. Firestore (≈ 340 KB) loads only on first use.

Optimizations applied

- **Server:** response cache with in-flight de-duplication (repeat requests made 0 upstream calls in testing); parallel image URL checks (worst case 6 s instead of 18 s per image); early stop on auth errors; per-attempt timeouts; about 450 lines of dead fallback data removed from `server.ts`.
- **HTTP:** hashed assets served with `Cache-Control: immutable, max-age=1y`; `index.html` revalidated on every load.
- **Client:** lazy tabs and export libraries; abortable requests (no stale state after topic changes); ECG animation pauses when the tab is hidden and respects reduced-motion settings; pair results cached in the drug matrix.
- **Usability:** removed page-wide `select-none` so learners can copy text; added missing Tailwind shades; mobile header no longer overflows.

10 Publish-readiness gate

Run `npm run verify` (typecheck, then build, then `check:publish`). The gate exits non-zero on any failure. `--static` skips the server boot; `--live` tests the real provider and fails any AI route that fell back to offline data. CI runs the offline gate on every pull request.

GROUP	CHECKS
Repository (6)	Required files present · .gitignore covers build output and secrets · no .env tracked · build/start/lint scripts · Node engine pinned · medical disclaimer in README
Secrets (2)	No credentials in source (OpenAI-style, AWS, GitHub, Slack, private keys, OAuth secrets; Google keys only in the Firebase config) · none in the server bundle
Build (3)	Production build present · initial JS within the 200 KB gzip budget · secondary tabs code-split
Runtime (22)	Server boots · health · status (no secrets leaked) · readiness · cases · analyze-symptoms (valid and invalid input) · drug interaction · topic drugs · generate-cases (cardiac and non-cardiac, no off-topic ST elevation) · mind map · symptom map · quiz · images · Sheets without auth → 401 · malformed JSON → 400 · unknown route → 404 · SPA shell · security headers · cache headers · rate limiter returns 429

Runtime readiness endpoint (`/api/readiness`)

Node ≥ 22.19 · NODE_ENV=production · frontend build present · AI provider configured · provider endpoint is HTTPS · backup provider · APP_URL is HTTPS · proxy trust · rate limiting · secrets from environment · memory headroom · cache statistics. The compliance tab adds three browser-side checks: HTTPS context, storage availability and live AI reachability.

11 Configuration reference

VARIABLE	DEFAULT	PURPOSE
`PORT`	3000	Listen port
`NODE_ENV`	(unset = dev)	`production` serves `dist/`; otherwise the Vite dev server
`APP_URL`	none	Public HTTPS origin (readiness check)
`TRUST_PROXY`	none	`1` behind Nginx or a load balancer
`GEMINI_API_KEY`	none (offline)	Provider key; never commit
`GEMINI_BASE_URL`	none	OpenAI-compatible endpoint (HTTPS)
`GEMINI_MODEL`	gemini-2.5-flash / gemini-3.5-flash	Primary model (direct / gateway)
`AI_FALLBACK_MODELS`	gpt-4o-mini	Comma-separated models tried in order
`AI_BACKUP_BASE_URL`, `AI_BACKUP_API_KEY`	none	Optional failover provider
`AI_TIMEOUT_MS`	45000	Total budget per AI call
`AI_ATTEMPT_TIMEOUT_MS`	28000	Budget per model attempt
`AI_MAX_CONCURRENCY`	4	Parallel upstream calls (keep below the provider limit)
`AI_CACHE_TTL_MS`, `AI_CACHE_MAX_ENTRIES`	1800000, 200	Response cache
`RATE_LIMIT_PER_MIN`	30	AI POST requests per client IP
`HTTPS_PROXY` / `HTTP_PROXY`	none	Outbound proxy, applied automatically
`VITE_FIREBASE_*`	firebase-applet-config.json	Per-deployment Firebase override (build time)

RECOMMENDED FOR AICREDITS.IN

```
GEMINI_BASE_URL=https://aicredits.in/v1
GEMINI_API_KEY=<rotated key>
GEMINI_MODEL=google/gemini-3.5-flash-lite
AI_FALLBACK_MODELS=openai/gpt-4.1-mini,gpt-4o-mini
AI_MAX_CONCURRENCY=4
TRUST_PROXY=1
```

12 Deployment

Local or VM

```
npm ci
npm run verify                       # typecheck + build + 33-check gate
npm run check:publish -- --live      # optional: real provider (uses credits)
npm prune --omit=dev
NODE_ENV=production pm2 start dist/server.cjs --name medmap
```

Docker

```
docker build -t medmap:prod .
docker run -d --name medmap -p 3000:3000 --env-file .env medmap:prod
curl http://localhost:3000/api/readiness
```

Reverse proxy and probes

- Terminate TLS at Nginx or a load balancer and proxy to port 3000 with `X-Forwarded-*` headers; set `TRUST_PROXY=1`.
- Proxy read timeout ≥ 60 s (chunked generation takes about 20 s, with headroom for retries).
- Liveness probe `/api/health`; readiness probe `/api/readiness`.
- The host must reach the AI provider, `commons.wikimedia.org`, `upload.wikimedia.org`, `creativecommons.org` and Google APIs.

CI pipeline (GitHub Actions, Node 22)

`npm ci` → `npm run lint` → `npm run build` → `npm run check:publish` → `npm audit --omit=dev --audit-level=high`.

13 Defects found and fixed

SEVERITY	DEFECT	FIX
Critical	Live provider key hard-coded in `server.ts`; it also made offline mode impossible	Removed; environment-only; secret scan in gate and CI
Critical	14 imported modules missing; app could not build	Implemented all of them
High	All AI calls failed behind a proxy (fetch ignores `HTTPS_PROXY`)	Proxy-aware fetch, loaded lazily
High	Large generations exceeded the gateway's 30 s cap	Chunked parallel generation with partial fallback
High	Bursts hit the 5-per-key concurrency limit	Semaphore and 429 back-off
High	Offline cases relabelled MI content for any topic	Topic-matched curated cases

SEVERITY	DEFECT	FIX
High	Image atlas always empty, then relied on invented URLs	Commons search with AI annotation and verification
High	Server crashed on Node 20 (undici 8 requires 22.19), including the Docker image	Lazy undici; Node 22 in CI, Docker and engines
Medium	Sync crashed on missing `outcome`; unvalidated spreadsheet ID; formula evaluation of cell values	Input clamping, ID regex, `RAW` input, 401 handling
Medium	Malformed AI output could crash the UI	Validation of cases, quiz, mind map; JSON repair
Medium	Off-topic images shown as "verified" offline	Return none when no curated match
Medium	Tier router reported "cloud" without a key and probed localhost from public pages	Uses `/api/status.configured`; Ollama probe opt-in
Low	Hard-coded port, HTML returned for unknown API routes, no JSON error handler, deploy guide build step that could not work	Env port, JSON 404 and errors, corrected docs
Low	New topic did not refresh the symptom map; misleading success message; reference to a non-existent chat tab	Auto-rebuild on apply; accurate messages

14 Verification evidence

TEST	RESULT
TypeScript typecheck and production build	**Pass**
Publish gate, offline mode	**33 / 33**
Publish gate, `--live` against aicredits.in	**All 8 AI content routes live** Images tested separately (Wikimedia throttled the test IP); Sheets grading needs a Google token, not run
GitHub Actions CI on PR #1 head `14cea8f`	**Success**
Node 20 crash reproduced on the old build; fixed build boots on Node 20 and 22	**Pass**
Browser end to end (Playwright), offline and live: every tab, case play-through, exports menu, journal, topic apply	**0 console errors**
Phone width 390 px	**No horizontal overflow**
Secret scanner catches a planted `sk-live-…` key	**Detected, gate fails**
Mock provider: array unwrap, case validation, cache hit, early stop on a bad key	**Pass**
Image pipeline (tuberculosis): real public-domain chest X-ray	**Verified, score 95**
Medical spot checks: DKA labs (pH 7.02, HCO_3 8, anion gap 24) and treatment; clarithromycin + simvastatin and sildenafil + nitroglycerin both flagged severe	**Clinically accurate**
Google sign-in and Sheets sync with a real account	**Not tested**
`docker build`	**Not run**
Image atlas across many topics from a dedicated IP	**Partial** Shared test IP throttled by Wikimedia

15 Known limitations and roadmap

- **Single-process state.** The rate limiter, AI cache and concurrency semaphore are per process. For multiple replicas, move them to Redis or enforce limits at the gateway.
- **AI content accuracy.** Output is model-generated and spot-checked, not clinically validated. A clinician review workflow, or citations to guideline sources, would be needed for formal teaching use.
- **Wikimedia dependence.** Image retrieval is subject to Wikimedia rate limits. An API token or a pre-built curated image index would remove this.
- **Logging.** Some legacy handlers still log plain text; migrate them to the JSON format.

- **Unused heavy dependencies.** `@huggingface/transformers` and `@mlc-ai/web-llm` are installed for the WebGPU tier but not yet wired to inference. Implement that tier or remove them to cut install size.
- **Tests.** Coverage is the end-to-end gate plus browser runs; unit tests for the chunk merge and validators would harden refactors.

16 Action items and change history

Owner actions

#	ACTION	WHY
1	Rotate both provider keys; store the new one only in the server environment or a secrets manager	Both keys were exposed
2	Switch the GitHub default branch to `main`; mark PR #1 ready and merge	The work branch became default because the repository was empty
3	Restrict the Firebase API key by HTTP referrer; add the production domain to Firebase Auth authorised domains	Prevents key misuse; sign-in fails on unlisted domains
4	On the production host run `npm run check:publish -- --live`, test Sheets sync with a real account, and run `docker build`	The three paths not yet verified
5	Plan Redis or gateway limits before scaling past one instance	Per-process state

Commits (PR #1)

COMMIT	SUMMARY
`f86168f`	Initial empty commit (base for review)
`927491c`	Complete the application: missing modules, publish gate, readiness endpoint, security, fixes, optimizations, audit 34 → 0
`f2a584e`	Make AI fully functional on the live provider: proxy support, chunking, concurrency control, model selection, real Commons images
`14cea8f`	Fix the Node 20 startup crash: lazy undici; runtime moved to Node 22

MedMap Visual Atlas application report, prepared 12 November 2026 from the repository at commit 14cea8f. Measurements come from test runs in a cloud development container against aicredits.in; latencies on other hosts and providers will differ. Educational use only; not medical advice.